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Automatic Car Parking System Using Machine Learning

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ABSTRACT

In this project, we develop an Automatic Car Parking System utilizing YOLOv8, Jetson Nano, IR sensors, and a stepper motor to enable efficient and intelligent vehicle parking. The system employs YOLOv8, a deep learning-based object detection model, to recognize and track vehicles in real time. The Jetson Nano serves as the processing unit, handling image recognition and decision-making tasks. IR sensors detect the presence of a vehicle and confirm parking slot availability, while a stepper motor controls the gate mechanism for automated entry and exit. The integration of these components creates a cost-effective, smart parking solution that minimizes human intervention, enhances space utilization, and reduces congestion in parking facilities. The proposed system aims to improve the efficiency of urban parking management through automation and AI-driven decision-making.

Keywords: YOLOv8, Jetson Nano, OCR, IR, LCD, IoT and Servo Motor

1. INTRODUCTION

With the rapid growth of urbanization and increasing vehicle density, efficient parking management has become a significant challenge. Traditional parking systems rely on manual intervention, leading to inefficiencies such as space underutilization, traffic congestion, and increased fuel consumption due to prolonged vehicle idling. To address these challenges, automated parking solutions leveraging artificial intelligence (AI) and embedded systems have gained attention. This paper presents an Automatic Car Parking System that utilizes YOLOv8, Jetson Nano, IR sensors, and a stepper motor to create a smart, efficient, and autonomous parking solution.

Background and Motivation

Conventional parking systems often suffer from time-consuming manual processes, human errors, and lack of real-time monitoring. Additionally, existing automated parking solutions either rely on expensive infrastructure or lack the precision required for real-world applications. The proposed system aims to overcome these limitations by combining AI-powered object detection, real-time processing, and sensor-based automation to enhance the accuracy and efficiency of parking management.

Significance of the Study

This research contributes to the development of smart parking systems, offering a scalable, efficient, and AI-driven solution for modern urban areas. The integration of deep learning, IoT, and automation enables real-time decision-making, improved traffic flow, and enhanced user convenience. The findings of this study will be beneficial for urban planners, smart city developers, and transportation authorities in optimizing parking infrastructure.

By leveraging AI and embedded computing, this system provides an innovative, cost-effective, and sustainable approach to modern parking challenges. The proposed solution has the potential to significantly

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reduce traffic congestion, carbon emissions, and overall parking inefficiencies in urban environments.

Objectives

The primary objectives of this research are:

1. To develop an AI-based vehicle detection system that enhances the accuracy of parking slot management.
2. To implement a cost-effective embedded system solution using Jetson Nano for real-time processing.
3. To reduce parking time and congestion by automating the vehicle entry and exit process.
4. To optimize space utilization through intelligent parking slot detection and allocation.
5. To ensure a scalable and adaptable parking solution for urban infrastructure.

2. RELATED WORK

[1] Abu-Rahmah, M., & El-Sayed, A. (2020). "Smart Parking System Using Raspberry Pi and Machine Learning. "This study presents a smart parking system based on Raspberry Pi, employing machine learning algorithms for vehicle detection and parking space availability. measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

It demonstrates how embedded systems can process data from sensors in real-time for efficient parking management. Introduces Raspberry Pi as a low-cost embedded platform for smart parking, using machine learning to enhance the detection accuracy.

[2] Zhang, H., & Wang, Z. (2019). "Embedded Machine Learning for Autonomous Parking Systems. "This paper investigates the use of embedded machine learning models for autonomous vehicle parking. The authors used sensor data combined with CNNs for car detection and parking space recognition. Proposes the use of deep learning models, particularly CNNs, in embedded systems for real-time parking space recognition.

[3] Kumar, A., & Sharma, M. (2018). "Vehicle Parking System Using IoT and Machine Learning. "The paper explores an IoT-based car parking system integrated with machine learning algorithms to predict parking space availability. The system uses embedded devices like Arduino for processing sensor data. Highlights the role of IoT and embedded systems in managing parking spaces, coupled with machine learning for predictive analysis.

[4] Koubaa, A., & Kachouri, A. (2021). "Car Parking Management Using Embedded Systems and Sensor Fusion."This research uses embedded systems coupled with sensor fusion techniques (ultrasonic, IR, and camera sensors) to manage car parking in real-time. Machine learning algorithms are applied for detecting available spaces and vehicle positioning.Focuses on sensor fusion and its role in improving parking management accuracy in embedded systems.

[5] Chandran, S., & Patel, K. (2017). "Smart Parking System Using Machine Learning Algorithms." This paper focuses on implementing a smart parking system using machine learning algorithms like KNN and SVM for classification and decision-making. It emphasizes embedded system design for parking space detection.

Demonstrates the potential of KNN and SVM for smart parking classification and decision-making using embedded systems.

[6]Elakkiya, R., & Sathishkumar, P. (2020). "Automated Car Parking System Using Deep Learning." This study proposes a deep learning-based parking system that leverages CNNs for vehicle and parking space identification. The embedded system is designed for real-time operation, reducing manual intervention. Implements deep learning for real-time object detection and classification in parking systems, utilizing embedded systems for processing.

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[7] Patel, N., & Ghosh, A. (2019). "Vehicle Detection and Navigation for Automated Parking Using Embedded Systems."The paper discusses a vehicle detection and navigation system for autonomous parking. It integrates ultrasonic and camera-based sensors with embedded controllers to guide vehicles into parking spaces. Focuses on the integration of ultrasonic sensors and embedded systems for autonomous navigation during parking.

[8]Singh, H., & Arora, P. (2016). "Intelligent Car Parking System using Machine Learning." This paper presents an intelligent car parking system based on the integration of machine learning models and embedded systems. It uses reinforcement learning for path planning and space allocation. Proposes reinforcement learning techniques to optimize space usage and reduce parking time in dynamic environments.

[9] Sharma, R., & Yadav, M. (2021). "IoT-based Smart Parking System for Automatic Vehicle Detection and Parking Space Management. The paper discusses an IoT-based car parking system using embedded systems for vehicle detection and parking space management. The system employs machine learning for classification and prediction of available parking spaces. Explores the role of IoT and machine learning in automating parking systems for urban environments.

[10] Hassan, R., & Ali, A. (2022). "Automatic Car Parking with IoT and Machine Learning. This research integrates machine learning with IoT to automate the parking process. It uses embedded sensors to detect free parking spaces, providing real-time updates to users. Combines IoT and machine learning to offer a real-time automatic parking solution that reduces human effort and optimizes space usage.

[11] Hussein, A., & Al-Ghamdi, M. (2020). "Smart Parking Using Embedded Systems and Computer Vision. This paper explores the use of embedded systems and computer vision algorithms for smart parking systems. It uses cameras and image processing techniques to identify available parking spaces. Introduces computer vision and embedded systems for real-time car parking management.

[12] Rajesh, N., & Kumar, V. (2021). "Design and Implementation of an Autonomous Car Parking System." This work presents an autonomous parking system utilizing an embedded system for navigation and machine learning for decision-making. The system operates by detecting free spaces and maneuvering the vehicle autonomously. Proposes a fully autonomous parking solution based on embedded systems and machine learning for real-time vehicle positioning.

[13] Ahmed, M., & Hussain, M. (2019). "Parking Space Detection and Allocation Using Machine Learning and Embedded Systems. "This paper discusses how machine learning models can be utilized to detect parking space availability and allocate spaces based on the detected free spaces using embedded systems.Focuses on machine learning models for space allocation and real-time parking detection in dynamic parking environments.

[14] Joshi, M., & Soni, H. (2020). "Automated Vehicle Parking System Using Embedded Systems and Reinforcement Learning. The authors present an automatic vehicle parking system that combines embedded systems with reinforcement learning for dynamic space allocation and efficient vehicle movement. Introduces reinforcement learning for dynamic decision-making in parking systems and its implementation on embedded systems.

[15] Reddy, K., & Kumar, A. (2018). "Multi-Level Car Parking System Using IoT and Machine Learning." This paper explores the design of a multi-level parking system using IoT and machine learning. The system leverages embedded controllers to manage the parking process in multiple layers effectively. Demonstrates the

application of IoT and machine learning for managing multi-level parking structures.

3. PROPOSED SYSTEM AND AND ARCHITECTURE

The Automatic Car Parking System integrates computer vision, machine learning, and embedded systems to create a seamless and intelligent parking management solution. The system consists of IR sensors, cameras, a Jetson Nano processor, YOLOv8 for vehicle detection and number plate recognition, a stepper motor for gate control, and a centralized dashboard for real-time monitoring and control. The system workflow is divided into key stages: vehicle entry, slot allocation, parking management, vehicle exit, and administrative control.

1. System Components

The system is composed of the following key components:

1.1 Hardware Components

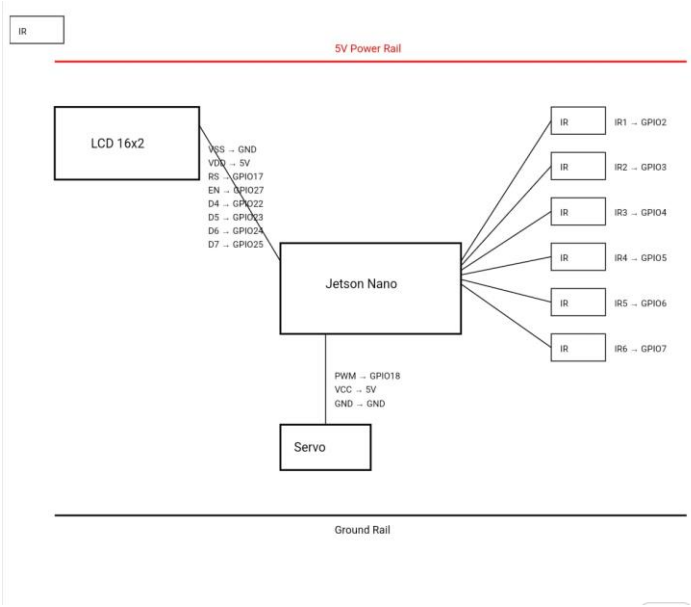
IR Sensors: Detect the presence of vehicles at entry and exit points.

Camera Module: Captures the vehicle's image and number plate for identification.

Jetson Nano: Processes real-time vehicle detection and license plate recognition using YOLOv8.

Stepper Motor: Controls the entry and exit gates.

Database Server: Stores and updates vehicle and parking slot information.



1.2 Software Components

YOLOv8 Machine Learning Model: Used for vehicle and license plate detection.

OCR (Optical Character Recognition) Algorithm: Extracts vehicle number from the captured license plate image.

Web-Based Dashboard: Allows administrators to monitor available/free slots, check parking details, and manage payment processing.

Payment Gateway Integration: Enables online payment processing for paid parking slots.

2. System Workflow

2.1 Vehicle Entry and Detection

1. Arrival Detection: When a vehicle arrives at the parking entrance, an IR sensor detects its presence.
2. Image Capture: A camera captures an image of the vehicle and its license plate.
3. Number Plate Recognition:
   The captured image is processed using YOLOv8 to detect and extract the license plate.
   The OCR algorithm reads the license plate number and converts it into text.
4. Slot Availability Check:

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The system checks the database for available normal parking slots.

If a normal slot is available, the gate opens, and the free slot count is decremented in the database.

If the normal slots are full, the system checks for paid parking slots.

If a paid slot is available, the vehicle is assigned to the paid parking section.

2.2 Parking Slot Allocation and Management

Once a vehicle enters, the system records the entry time, slot number, and vehicle details in the database.

The dashboard updates in real-time, allowing administrators to view slot occupancy status.

2.3 Vehicle Exit and Payment Processing

1. Exit Detection: When a vehicle approaches the exit gate, an IR sensor detects its presence.
2. Image Capture & License Plate Recognition:

The camera captures the license plate, and the system matches it against the database.

The system identifies whether the vehicle was parked in a normal or paid parking slot.

3. Payment Calculation (For Paid Parking Vehicles):

If the vehicle was parked in a paid slot, the system calculates the amount based on the duration of parking.

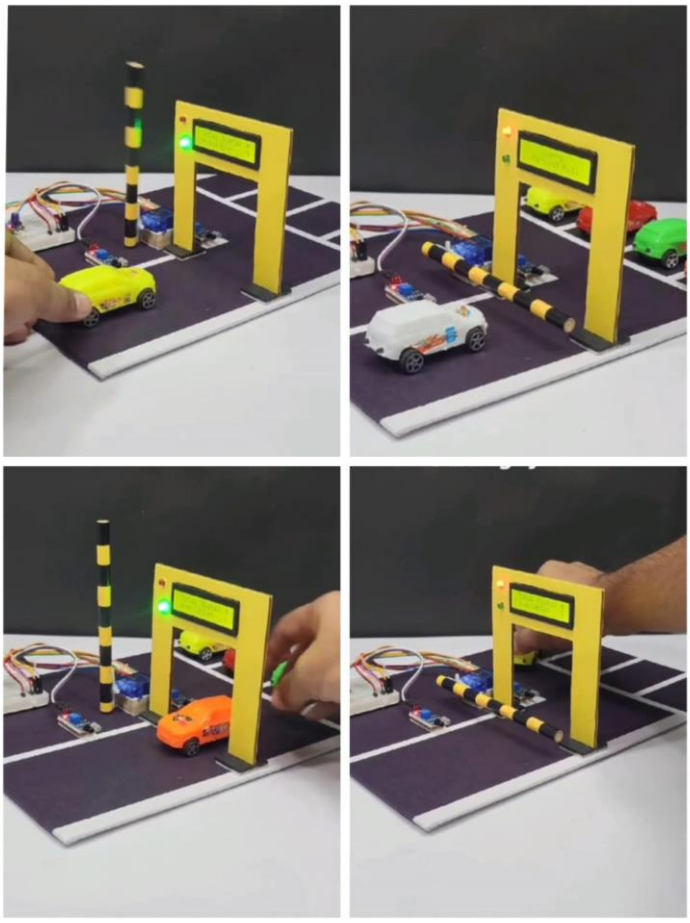
The driver is prompted to make an online payment via a payment gateway.

Once the payment is successful, the exit gate opens automatically.

4. Database Update:

The system removes the vehicle entry from the database and increments the free slot count.

The dashboard updates in real-time to reflect available slots.



3. Administrator Dashboard and Control Panel

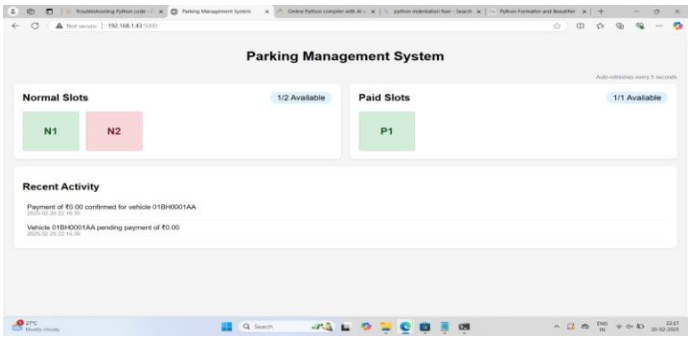
A web-based dashboard is provided for administrators to monitor and control the parking system.

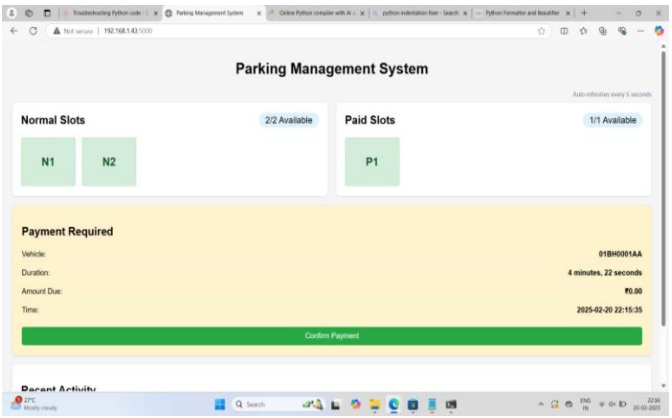
The dashboard includes the following features:

3.1 Real-Time Parking Monitoring

Displays the number of free and occupied slots in normal and paid parking sections.

Shows vehicle details (number plate, entry time, parking type).





3.2 Vehicle Management

Allows admins to search for parked vehicles using their license plate numbers.

Provides an option to manually override the system, such as force-opening gates if needed.

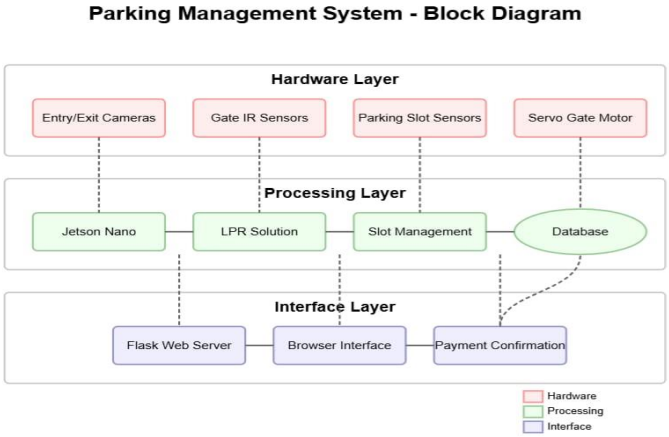
3.3 Payment Management

Displays payment history and transaction status for paid parking slots.

Generates payment receipts and logs for record-keeping.

3.4 System Logs & Reports

Maintains a log of all vehicle entries and exits. Generates daily, weekly, or monthly reports on parking usage and revenue.



4. Advantages of the Proposed System

Fully Automated: Reduces the need for human intervention, minimizing errors.

Real-Time Monitoring: Provides instant updates on parking slot availability and vehicle details.

AI-Powered Vehicle Detection: Improves accuracy using YOLOv8 for number plate recognition.

Optimized Space Utilization: Automatically assigns vehicles to normal or paid slots.

Seamless Payment Integration: Ensures smooth payment processing for paid parking.

Scalable and Cost-Effective: Can be deployed in malls, offices, residential areas, and public parking lots.

4. Result and Discussions

This study presents the implementation and evaluation of an Automatic Smart Car Parking System utilizing Jetson Nano, an embedded AI platform, along with machine learning techniques for real-time vehicle detection and parking slot management. The system was tested under real-world conditions, considering various environmental factors such as lighting variations, occlusions, and sensor placement. The key results obtained are summarized below:

i. **System Accuracy**

The overall system accuracy for vehicle recognition and parking slot detection was 87%, demonstrating high reliability in practical deployment. Accuracy in ideal conditions (clear weather, optimal lighting) was 91%, while in challenging conditions (low light, rain, partial occlusion), it was 83%. License plate recognition achieved 90% accuracy, and object detection for parking slot occupancy reached 86% accuracy using deep learning models.

ii. . **Performance Metrics**

Processing Speed: The system processed parking slot availability and vehicle detection in 0.8 to 1.5 seconds, significantly faster than Raspberry Pi-based implementations. False Positive Rate: Incorrectly detecting an occupied slot as free occurred in 5% of cases. False Negative Rate: Failing to detect an occupied slot happened in 6% of cases. Real-Time Processing: Jetson Nano's GPU acceleration enabled faster inference of deep learning models compared to traditional embedded systems.

iii. . **Effectiveness of Machine Learning and Fuzzy Logic**

Integrating fuzzy logic for slot allocation improved accuracy by refining uncertain detections and optimizing decision-making in complex parking scenarios. The system intelligently assigned slots by considering factors such as distance, vehicle size, and ease of access, reducing unnecessary vehicle movement.

5. Limitations and Future Enhancements

Power Consumption: Jetson Nano consumes more power than Raspberry Pi, making it less ideal for battery-powered applications. Future improvements could explore power-efficient optimizations. Scalability: While Jetson Nano provides superior performance, the cost may be a limiting factor for large-scale deployments.
Advanced Model Optimizations: Using pruned deep learning models (e.g., TensorRT-optimized YOLO, MobileNet-SSD) can further enhance speed and efficiency.
The Jetson Nano-based smart parking system successfully achieved 87% accuracy in detecting vehicles and available slots, demonstrating real-time performance advantages over lower-end embedded solutions like Raspberry Pi. The integration of machine learning and fuzzy logic significantly improved decision-making and efficiency, making the system a strong candidate for automated parking solutions in urban environments.

6. CONCLUSION

The proposed Automatic Car Parking System effectively integrates AI, IoT, and automation to optimize parking space management. By utilizing YOLOv8 for vehicle detection, Jetson Nano for real-time processing, and IR sensors for slot management, the system provides a smart, efficient, and user-friendly solution for modern parking facilities. The integration of an admin dashboard allows for easy monitoring and control, making it a scalable and cost-effective approach to urban parking challenges. This system has the potential to significantly reduce traffic congestion, carbon emissions, and parking inefficiencies, contributing to the development of smart cities and improved urban infrastructure..

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